Lecture 2

Exercise 1

1 Preliminary

Note that $V_t = V(t)$ in Lecture 2.

Consider the following stochastic differential equation

$$\begin{cases} dV_t = rV_t dt + \sigma V_t dW_t, \\ V_0 = 1, \end{cases} \tag{1}$$

where r and σ are constant.

Let $f(t, V_t) = \ln V_t$. Then, we get

$$\frac{\partial f}{\partial t} = 0, \quad \frac{\partial f}{\partial V_t} = \frac{1}{V_t}, \quad \frac{\partial^2 f}{\partial V_t^2} = -\frac{1}{V_t^2}.$$

Applying Itô Lemma yields

$$\begin{aligned} d\left(\ln V_t\right) &= d\left(f(t, V_t)\right) \\ &= \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial V_t} r V_t + \frac{1}{2}\frac{\partial^2 f}{\partial V_t^2}\sigma^2 V_t^2\right) dt + \frac{\partial f}{\partial V_t}\sigma V_t dW_t \\ &= \left\{0 + \frac{1}{V_t} r V_t + \frac{1}{2}\Big(-\frac{1}{V_t^2}\Big)\sigma^2 V_t^2\right\} dt + \frac{1}{V_t}\sigma V_t dW_t \\ &= \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t. \end{aligned} \tag{2}$$

Hence,

$$\ln V_T - \ln V_t = \int_t^T \left(r - \frac{1}{2}\sigma^2\right) ds + \int_t^T \sigma dW_s = \left(r - \frac{1}{2}\sigma^2\right)(T - t) + \sigma(W_T - W_t),$$

i.e.,

$$\ln\left(\frac{V_T}{V_t}\right) = \left(r - \frac{1}{2}\sigma^2\right)(T - t) + \sigma(W_T - W_t),$$

i.e.,

$$V_T = V_t \exp\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma(W_T - W_t)\right). \tag{3}$$

Let $X_t = \ln V_t$. Then, it follows from (2) that we have

$$dX_t = \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t. \tag{4}$$

Taking the expectation for (4) leads to

$$d(\mathbb{E}^*[X_t]) = \left(r - \frac{1}{2}\sigma^2\right) dt, \tag{5}$$

since $\mathbb{E}^*[dW_t] = 0$. Hence,

$$\mathbb{E}^*[X_T] - \mathbb{E}^*[X_t] = \int_t^T \left(r - \frac{1}{2}\sigma^2\right) ds = \left(r - \frac{1}{2}\sigma^2\right)(T-t). \tag{6}$$

Let $Y_t = X_t - \mathbb{E}[X_t]$. Then

$$\begin{aligned} dY_t &= X_t - \mathbb{E}^*[X_t] \\ &= \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t - \left(r - \frac{1}{2}\sigma^2\right) dt \\ &= \sigma dW_t. \end{aligned} \tag{7}$$

Let $g(t, Y_t) = Y_t^2$. Then, we get

$$\frac{\partial g}{\partial t} = 0, \quad \frac{\partial g}{\partial V_t} = 2Y_t, \quad \frac{\partial^2 g}{\partial V_t^2} = 2.$$

Applying Itô Lemma yields

$$\begin{aligned} d\left(Y_t^2\right) &= d\left(g(t, V_t)\right) \\ &= \left(\frac{\partial g}{\partial t} + \frac{\partial g}{\partial Y_t} \times 0 + \frac{1}{2}\frac{\partial^2 g}{\partial Y_t^2}\sigma^2\right) dt + \frac{\partial f}{\partial Y_t}\sigma dW_t \\ &= \left\{0 + 2Y_t \times 0 + \frac{1}{2} \times 2 \times \sigma^2\right\} dt + 2Y_t \sigma dW_t \\ &= \sigma^2 dt + 2\sigma Y_t dW_t. \end{aligned} \tag{8}$$

Taking the expectation for (8) leads to

$$d(\mathbb{E}^*[Y_t^2]) = \sigma^2 dt, \tag{9}$$

since $\mathbb{E}^*[dW_t] = 0$. Hence,

$$\mathbb{E}^*[Y_T^2] - \mathbb{E}^*[Y_t^2] = \int_t^T \sigma^2 ds = \sigma^2(T-t). \tag{10}$$

Further, we get

$$\begin{aligned}\mathrm{Var}(X_T) - \mathrm{Var}(X_t) &= \mathbb{E}^*[(X_T - \mathbb{E}^*[X_T])^2] - \mathbb{E}^*[(X_t - \mathbb{E}^*[X_t])^2] \\ &= \mathbb{E}^*[Y_T^2] - \mathbb{E}[Y_t^2] \\ &= \sigma^2(T-t).\end{aligned} \tag{11}$$

Therefore, we have

$$X_T - X_t \sim N\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t), \sigma^2(T-t)\right),$$

i.e.,

$$\ln V_T - \ln V_t \sim N\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t), \sigma^2(T-t)\right), \tag{12}$$

i.e.,

$$\ln\left(\frac{V_T}{V_t}\right) \sim N\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t), \sigma^2(T-t)\right),$$

i.e.,

$$\ln\left(\frac{V_T}{V_t}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t) \sim N\left(0, \sigma^2(T-t)\right),$$

i.e.,

$$Z := \frac{\ln\left(\frac{V_T}{V_t}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \sim N(0,1).$$

Hence, using the standard normal variable Z, we re-express V_T as

$$V_T = V_t \exp\left(\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma\sqrt{T-t}Z\right). \tag{13}$$

$$\sigma\sqrt{T-t}\,Z = \ln\left(\frac{V_T}{V_t}\right) - \left(r - \frac{1}{2}\sigma^2\right)(T-t)$$

$$\left(r - \frac{1}{2}\sigma^2\right)(T-t) + \sigma\sqrt{T-t}\,Z = \ln\left(\frac{V_T}{V_t}\right)$$

$$\exp(\quad) = \frac{V_T}{V_t}$$

2 Black-Scholes-Merton Geometric Brownian Models

Now let us consider

$$\begin{aligned} P^*(V_T \le D) &= P^*\left(\frac{V_T}{V_t} \le \frac{D}{V_t}\right) \\ &= P^*\left\{\ln\left(\frac{V_T}{V_t}\right) \le \ln\left(\frac{D}{V_t}\right)\right\} \\ &= P^*\left\{\ln\left(\frac{V_T}{V_t}\right) - \left(r-\frac{1}{2}\sigma^2\right)(T-t) \le \ln\left(\frac{D}{V_t}\right) - \left(r-\frac{1}{2}\sigma^2\right)(T-t)\right\} \\ &= P^*\left\{\frac{\ln\left(\frac{V_T}{V_t}\right) - \left(r-\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}} \le \frac{\ln\left(\frac{D}{V_t}\right) - \left(r-\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}\right\} \\ &= P(Z \le -d_2) \\ &= \Phi(-d_2), \end{aligned} \tag{14}$$

where $d_2 = -\frac{\ln\left(\frac{D}{V_t}\right) - \left(r-\frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}$.

At the beginning $t = 0$, we have

$$\begin{aligned} P^*(V_T \le D) &= \Phi\left(\frac{\ln\left(\frac{D}{V_0}\right) - \left(r-\frac{1}{2}\sigma^2\right)(T-0)}{\sigma\sqrt{T-0}}\right) \\ &= \Phi\left(\frac{\ln D - \left(r-\frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right). \end{aligned} \tag{15}$$

Hence, the tax credit can be presented by

$$\begin{aligned} A(0) &= \mathbb{E}^*[e^{-rT}\kappa D \cdot I\{V(T) \ge D\}] \\ &= e^{-rT}\kappa D \cdot \mathbb{E}^*[I\{V(T) \ge D\}] \\ &= e^{-rT}\kappa D \cdot P^*(V(T) \ge D) \\ &= e^{-rT}\kappa D \cdot P^*(V_T \ge D) \\ &= e^{-rT}\kappa D\,(1 - P^*(V_T \le D)) \\ &= e^{-rT}\kappa D\left(1 - \Phi\left(\frac{\ln V_T - \left(r-\frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right)\right). \end{aligned} \tag{16}$$

It follows from (12) that we have

$$\ln V_T \sim N\left(\left(r-\frac{1}{2}\sigma^2\right)T, \sigma^2 T\right), \tag{17}$$

for $t = 0$. Then, we have its density

$$p^*(V_T) = \frac{1}{\sqrt{2\pi T}\sigma V_T} \exp\left(-\frac{\left(\ln V_T - \left(r - \frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2 T}\right). \tag{18}$$

The bankruptcy cost is given by

$$\begin{aligned}
C(0) &= \mathbb{E}^*[e^{-rT}(1-\alpha)V(T)\cdot I\{V(T)\le D\}] \\
&= e^{-rT}(1-\alpha)\mathbb{E}^*[V(T)\cdot I\{V(T)\le D\}] \\
&= e^{-rT}(1-\alpha)\mathbb{E}^*[V_T\cdot I\{V_T\le D\}] \\
&= e^{-rT}(1-\alpha)\int_{-\infty}^{D} V_T\cdot p^*(V_T)dV_T \\
&= e^{-rT}(1-\alpha)\int_{-\infty}^{D} V_T\cdot\frac{1}{\sqrt{2\pi T}\sigma V_T}\exp\left(-\frac{\left(\ln V_T-\left(r-\frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2T}\right)dV_T \\
&= e^{-rT}(1-\alpha)\int_{-\infty}^{D} V_T\cdot\frac{1}{\sqrt{2\pi T}\sigma}\exp\left(-\frac{\left(\ln V_T-\left(r-\frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2T}\right)d(\ln V_T) \\
&= e^{-rT}(1-\alpha)\int_{-\infty}^{D} \exp(\ln V_T)\cdot\frac{1}{\sqrt{2\pi T}\sigma}\exp\left(-\frac{\left(\ln V_T-\left(r-\frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2T}\right)d(\ln V_T) \\
&= e^{-rT}(1-\alpha)\int_{-\infty}^{D} \frac{1}{\sqrt{2\pi T}\sigma}\exp\left(-\frac{\left(\ln V_T-\left(r-\frac{1}{2}\sigma^2\right)T\right)^2-2\sigma^2T\ln V_T}{2\sigma^2T}\right)d(\ln V_T) \\
&= e^{-rT}(1-\alpha)\int_{-\infty}^{D} \frac{1}{\sqrt{2\pi T}\sigma}\exp\left(-\frac{\left(\ln V_T-\left(r+\frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2T}\right)\exp(rT)d(\ln V_T) \\
&= (1-\alpha)\int_{-\infty}^{D} \frac{1}{\sqrt{2\pi T}\sigma}\exp\left(-\frac{\left(\ln V_T-\left(r+\frac{1}{2}\sigma^2\right)T\right)^2}{2\sigma^2T}\right)d(\ln V_T) \\
&= (1-\alpha)\Phi\left(\frac{\ln D-\left(r+\frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}\right).
\end{aligned} \tag{19}$$

$e^a e^b = e^{a+b}$